2.(iv) $x + y\frac{dy}{dx} = 0$ 3. $r^2\left(\frac{d^2y}{dx^2}\right)^2 = \left[1+\left(\frac{dy}{dx}\right)^2\right]^3$ 4. $y = x\frac{dy}{dx}$

5. $2a\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = 0$ 6. $y^2 = x^2 + 2xy\frac{dy}{dx}$ 7. $y = 2x\frac{dy}{dx} + y\left(\frac{dy}{dx}\right)^2$

பயிற்சி 4.2

1.(i) $e^{-y} + ax + c = 0$ (ii) $\log x + \frac{x^2}{2} = \frac{y^2}{2} + \frac{y^3}{3} + c$ 2. $\log x - x = \log y + c$

3.(i) $x = cy$ (ii) $\log(1+y) = -e^x + c$ 4. $(1+\sin x) = c(1+\cos y)$

5. $(x-1)(y+1) = c$ 6.(i) $\log y = \frac{-\cos 2x}{2} + c$ (ii) $\frac{e^{ax}}{a} = \frac{-e^{by}}{b} + c$

7. $(y-b)^2 = (x-a)^2 + b^2 - a^2$

பயிற்சி 4.3

1. $x = ce^{\frac{y}{x}}$ 2. $x + y = ke^{\frac{-2x}{x+y}}$ 3. $y + \sqrt{x^2+y^2} = x^2c$

4. $3y^2 - 4yx + 3x^2 = x^3c$ 5. $\left(xy - y^2\right)x = c$ 6. $y\sqrt{y^2-x^2} = 2\sqrt{3}\,x^5$ 7. $y = ce^{\frac{x^2}{2y^2}}$

பயிற்சி 4.4

1. $\frac{y}{x} = x + c$ 2. $ye^{\sin x} = e^{\sin x}(\sin x - 1) + c$ 3. $x^2y = \frac{x^6}{6} + c$

4. $y\left(1+x^3\right) = x + \frac{x^3}{3} + c$ 5. $xy = e^x\left(x^2 - 2x + 2\right) + c$ 6. $y\sec x = \frac{1}{2}\left(x + \frac{\sin 2x}{2}\right) + c$

7. $y\sec^2 x = \sec x - 2$ 8. $x^2e^x - 2xe^x + 2e^x + c$ 9. ₹ 2,22,550

பயிற்சி 4.5

1. $y = Ae^{2x} + Be^{4x}$ 2. $y = (Ax+B)e^{2x}$ 3. $y = e^{-x}(A\cos\sqrt{2}\,x + B\sin\sqrt{2}\,x)$

4. $y = (Ax+B)e^{kx}$ 5. $y = \frac{e^{-3x}}{12} + \frac{e^{5x}}{20}$ 6. $y = Ae^{\frac{1}{2}x} + Be^{\frac{-3}{2}x} + \frac{e^{2x}}{21}$

7. $y = A\cos 4x + B\sin 4x$ 8. $y = e^x - \frac{3}{2}e^{2x} + \frac{e^{3x}}{2}$

9. $y = Ae^{-3x} + Be^{2x} + \frac{e^{3x}}{6} - \frac{x}{5}e^{-3x}$ 10. $y = (Ax+B)e^{5x} + 2x^2e^{5x} + \frac{1}{5}$

11. $y = Ae^{\frac{-3}{2}x} + Be^{\frac{-5}{2}x} + xe^{-\frac{3}{2}x}$ 12. $y = Ae^{2x} + Be^{\frac{-7}{3}x} + xe^{2x}$ 13. $p = Ae^{-4t} + Be^{2t} + 2$

பயிற்சி 4.6

1	2	3	4	5	6	7	8	9	10	11	12	13
(a)	(d)	(a)	(b)	(a)	(a)	(d)	(c)	(a)	(c)	(b)	(a)	(b)
14	15	16	17	18	19	20	21	22	23	24	25	
(c)	(d)	(a)	(d)	(a)	(b)	(d)	(a)	(a)	(d)	(c)	(a)	

XII Std - Business Maths & Stat TM Answers.indd 283 3/5/2020 1:28:43 PM